

AMARILLO NWS, TX, USA

WMO: 723630

Lat: 35.2331N

Lon: 101.7092W

Elev: 3593

StdP: 12.89

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	16.0	-0.7	6.0	26.3	2.9	7.3	32.7	32.9	47.8	29.8	49.7	11.3	0	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	98.8	65.6	96.1	65.9	93.6	66.1	71.4	86.7	70.3	86.0	69.2	85.2	14.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	114.7	75.5	66.1	110.1	74.6	65.0	105.8	73.9	37.5	86.5	36.4	85.7	35.5	85.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.4	26.8	24.5	DB	3.7	103.0	4.6	3.3	0.4	105.3	-2.2	107.3	-4.8	109.1	-8.1	111.5
			WB	2.7	73.2	4.3	1.4	-0.4	74.2	-2.9	75.1	-5.3	75.9	-8.5	76.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	38.3	41.2	49.2	56.9	66.2	75.6	79.0	77.6	70.5	58.8	47.5	38.5
	DBStd	17.00	9.73	10.98	10.18	9.08	8.34	6.12	4.75	5.03	7.06	8.78	9.60	9.76
	HDD50	1388	375	279	138	40	3	0	0	0	1	26	155	371
	HDD65	3952	827	667	495	265	87	4	0	1	31	226	526	823
	CDD50	4439	13	33	112	246	506	767	899	855	615	300	80	13
	CDD65	1526	0	0	4	21	125	321	434	391	195	34	1	0
	CDH74	18308	2	26	172	529	1737	3769	5197	4257	2046	507	65	1
	CDH80	9006	0	4	38	161	786	1951	2804	2206	896	154	6	0
Wind	WSAvg	12.9	12.2	12.9	13.8	14.9	13.7	13.9	12.2	11.4	12.0	12.6	12.7	12.2
Precipitation	PrecAvg	20.0	0.6	0.6	1.1	1.2	2.4	3.4	2.8	3.1	1.9	1.5	0.7	0.6
	PrecMax	36.7	3.1	2.5	4.1	6.5	9.3	10.7	8.0	9.1	5.0	6.5	4.1	2.5
	PrecMin	7.2	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.3	0.0	0.0	0.0	0.0
	PrecStd	5.3	0.7	0.6	1.1	1.3	1.7	2.3	1.9	2.0	1.3	1.4	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.8	77.5	85.0	89.2	97.4	103.2	101.9	100.5	96.9	89.1	80.4	71.3
		MCWB	48.5	50.9	53.7	55.3	60.7	63.4	66.6	66.6	65.2	60.6	52.8	48.0
	2%	DB	66.6	71.8	78.8	84.9	93.1	98.4	98.3	96.9	92.7	84.2	75.2	65.7
		MCWB	46.0	48.5	52.3	54.6	60.0	63.8	67.1	66.9	64.6	59.6	52.6	46.2
	5%	DB	60.9	66.5	73.9	80.2	88.8	94.4	95.7	94.1	89.2	80.2	69.8	60.6
		MCWB	43.7	46.7	50.9	54.5	60.0	65.4	67.3	67.3	64.9	58.7	50.2	44.5
	10%	DB	55.4	60.3	68.5	75.7	84.0	90.8	92.9	91.4	85.3	75.0	64.5	55.3
		MCWB	41.5	43.9	48.8	53.6	59.8	65.6	67.5	67.1	64.3	56.9	48.4	42.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	53.4	59.4	63.3	68.5	72.4	73.2	72.4	70.3	67.0	58.3	52.5
		MCDB	67.2	68.2	69.2	73.8	81.3	86.6	87.2	87.4	84.8	75.1	67.3	62.7
	2%	WB	47.2	50.3	55.8	60.1	66.4	70.5	71.4	71.1	68.7	64.4	55.2	48.2
		MCDB	63.1	67.7	69.4	72.0	80.2	85.7	86.9	86.6	82.9	73.7	66.0	62.5
	5%	WB	44.5	47.8	53.1	57.8	64.8	69.2	70.2	69.9	67.4	62.3	52.7	45.4
		MCDB	59.2	64.5	69.0	71.2	78.7	84.8	86.7	85.6	81.5	72.0	65.3	59.4
	10%	WB	41.8	44.9	50.5	55.4	63.0	68.1	69.1	69.0	66.2	59.9	50.2	42.3
		MCDB	54.7	59.6	66.1	71.0	77.1	83.5	86.1	84.4	79.7	70.2	63.5	54.2
Mean Daily Temperature Range	5% DB	MDBR	26.0	26.8	28.1	27.9	26.5	25.7	24.9	24.1	24.4	25.6	26.1	24.5
		MCDBR	34.8	36.3	35.9	35.1	33.4	30.9	28.6	27.8	29.4	32.1	33.8	33.2
		MCWBR	18.8	18.5	16.2	14.8	12.9	9.6	8.0	8.0	9.7	13.2	16.2	18.0
	5% WB	MCDBR	32.1	33.2	31.7	28.6	27.5	25.2	24.4	23.7	23.9	23.6	28.6	31.2
		MCWBR	17.8	18.0	15.7	14.8	12.7	9.6	8.4	8.0	9.0	12.3	15.1	17.8
Clear-Sky Solar Irradiance	taub	0.253	0.261	0.290	0.326	0.338	0.378	0.381	0.370	0.340	0.299	0.271	0.256	
	taud	2.603	2.589	2.511	2.409	2.438	2.372	2.403	2.400	2.458	2.564	2.570	2.590	
	Ebn at Noon	309	317	313	304	299	286	285	286	291	298	298	301	
	Edh at Noon	23	27	31	37	36	39	37	37	33	27	24	22	
All-Sky Solar Radiation	RadAvg	996	1253	1649	2031	2248	2390	2306	2065	1762	1348	1052	858	
	RadStd	85	113	118	123	171	143	98	101	93	146	103	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (20 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	N/A
		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	N/A

Nomenclature: See separate page